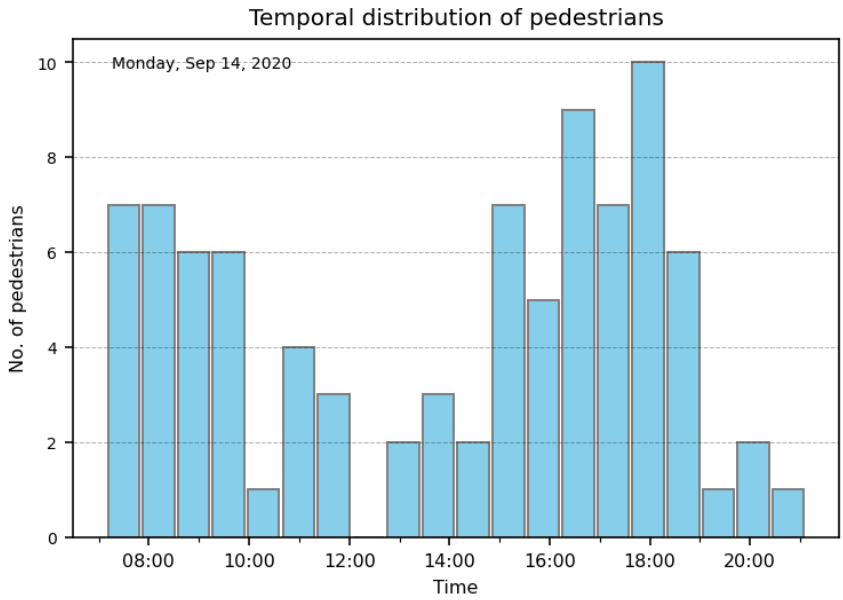
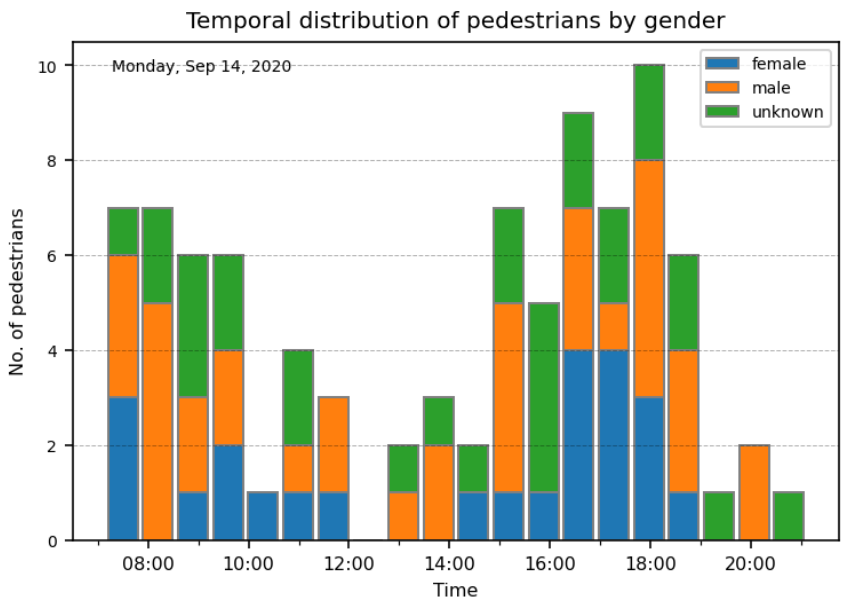
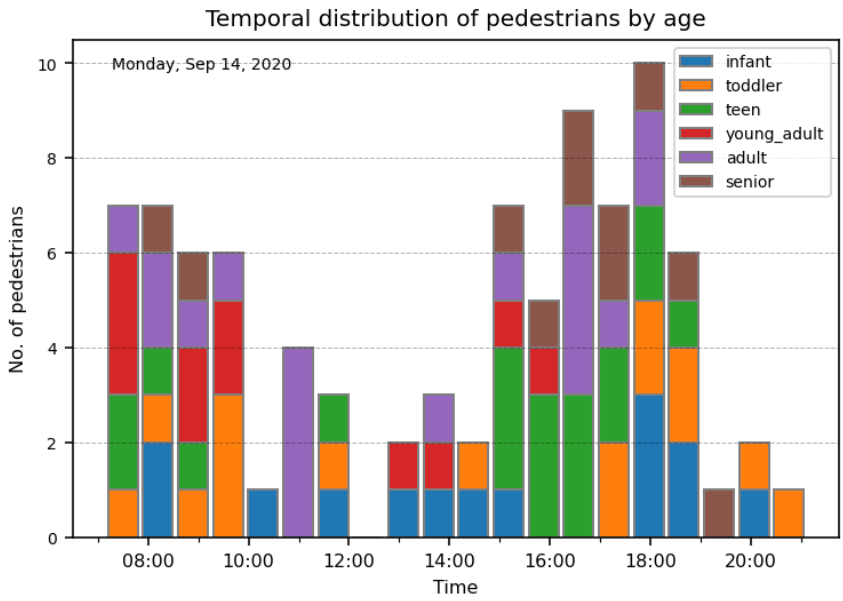
Derived indicators

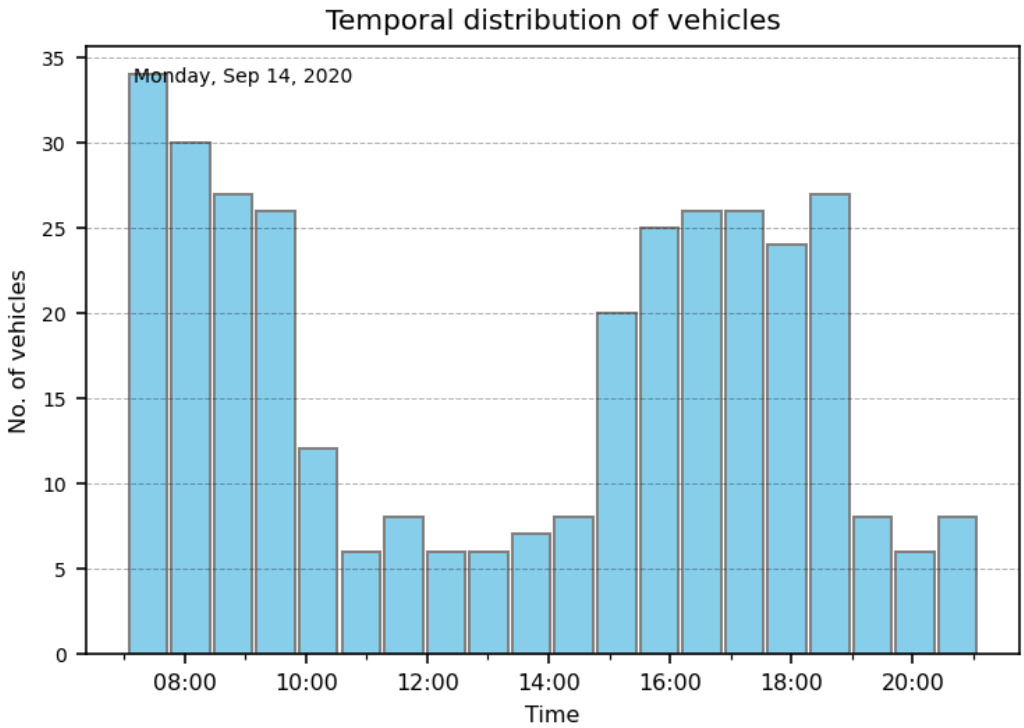
In order to analyze the simulated dataset, a couple of functions are developed to derive some indicators from the dataset. In this stage the plot of temporal distribution of street user movements, the histogram of their characteristics, the histogram of pedestrians' activity types and also the OD matrix of all street users are developed. The function of speed histogram for vehicles and bikes is not completed yet.

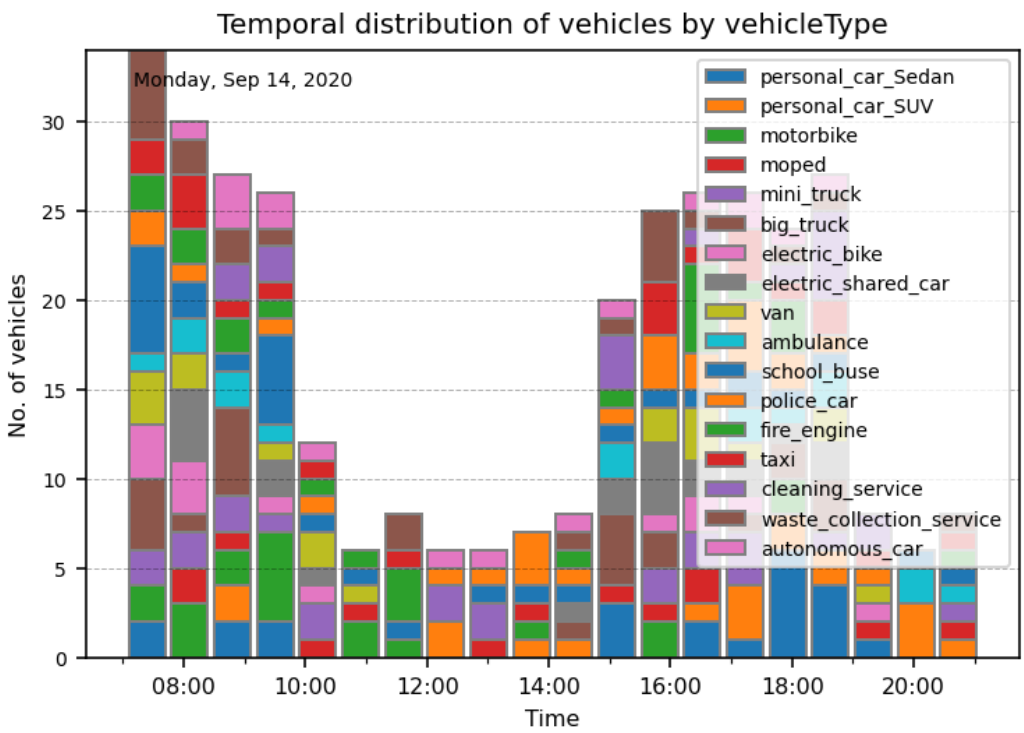
In the following figures, the results of the developed functions are shown.

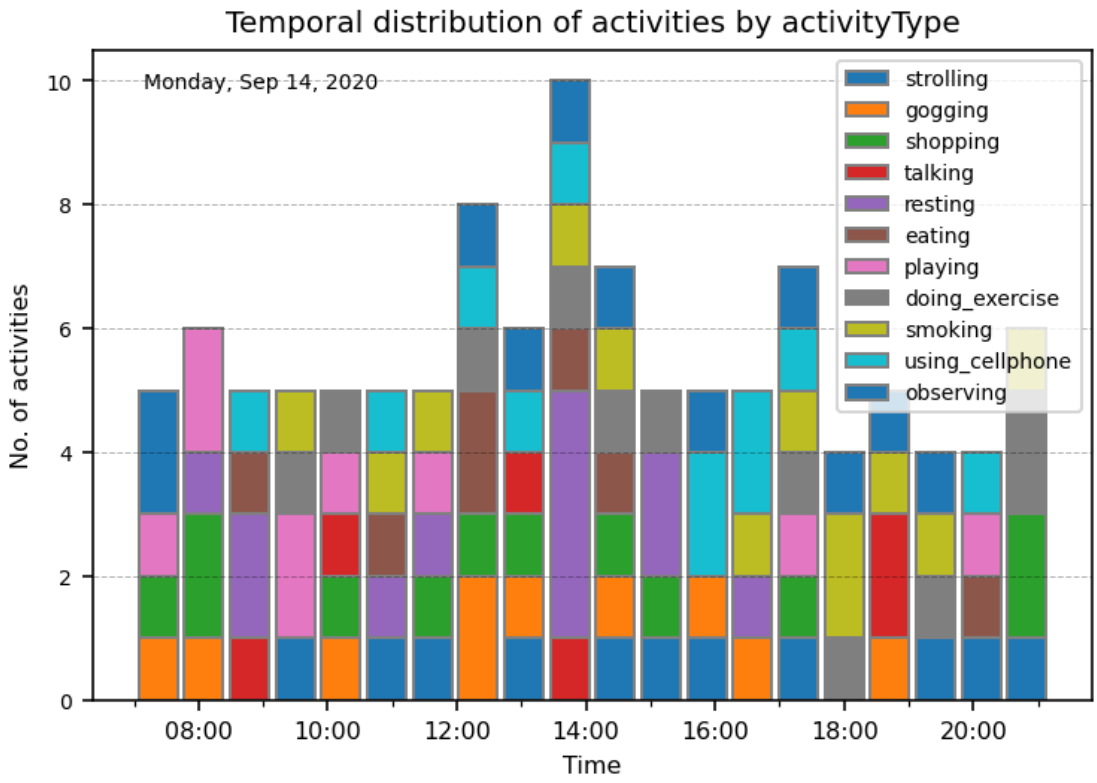


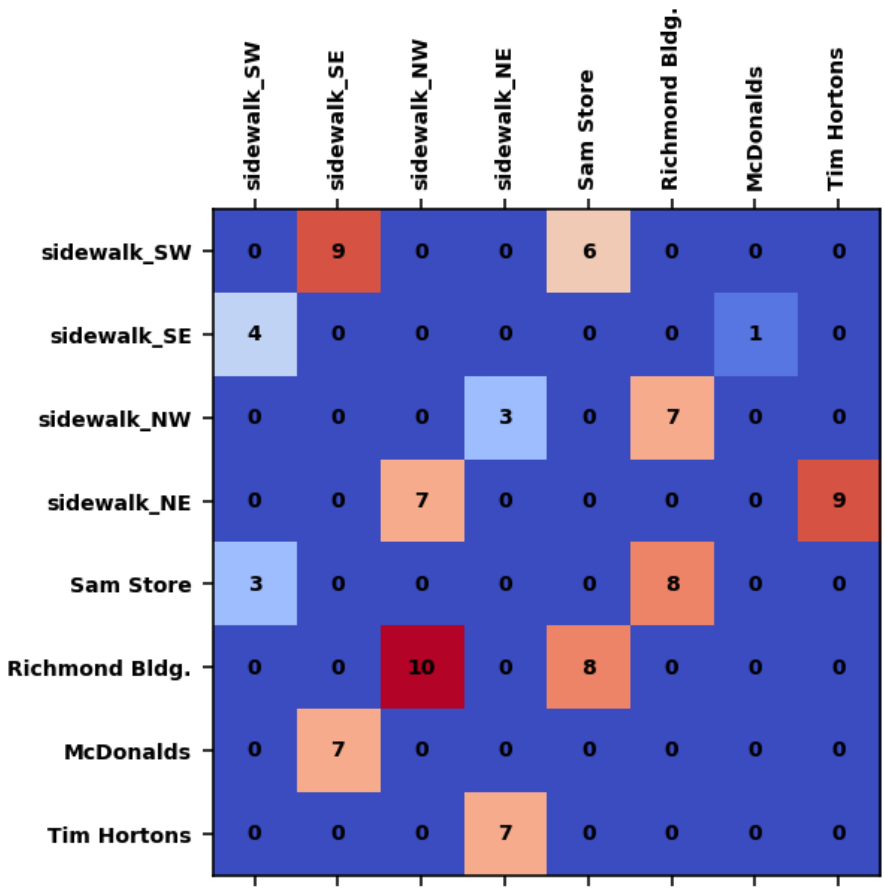



Temporal distribution of pedestrians by age
Monday, Sep 14, 2020
infant
toddler
teen
young_adult
adult
senior
No. of pedestrians
0
2
4
6
8
10
08:00
10:00
12:00
14:00
16:00
18:00
20:00
Time


Temporal distribution of vehicles
Monday, Sep 14, 2020
No. of vehicles
0
5
10
15
20
25
30
35
08:00
10:00
12:00
14:00
16:00
18:00
20:00
Time


Temporal distribution of vehicles by vehicleType
Monday, Sep 14, 2020
No. of vehicles
Time
0
5
10
15
20
25
30
08:00
10:00
12:00
14:00
16:00
18:00
20:00
personal_car_Sedan
personal_car_SUV
motorbike
moped
mini_truck
big_truck
electric_bike
electric_shared_car
van
ambulance
school_buse
police_car
fire_engine
taxi
cleaning_service
waste_collection_service
autonomous_car

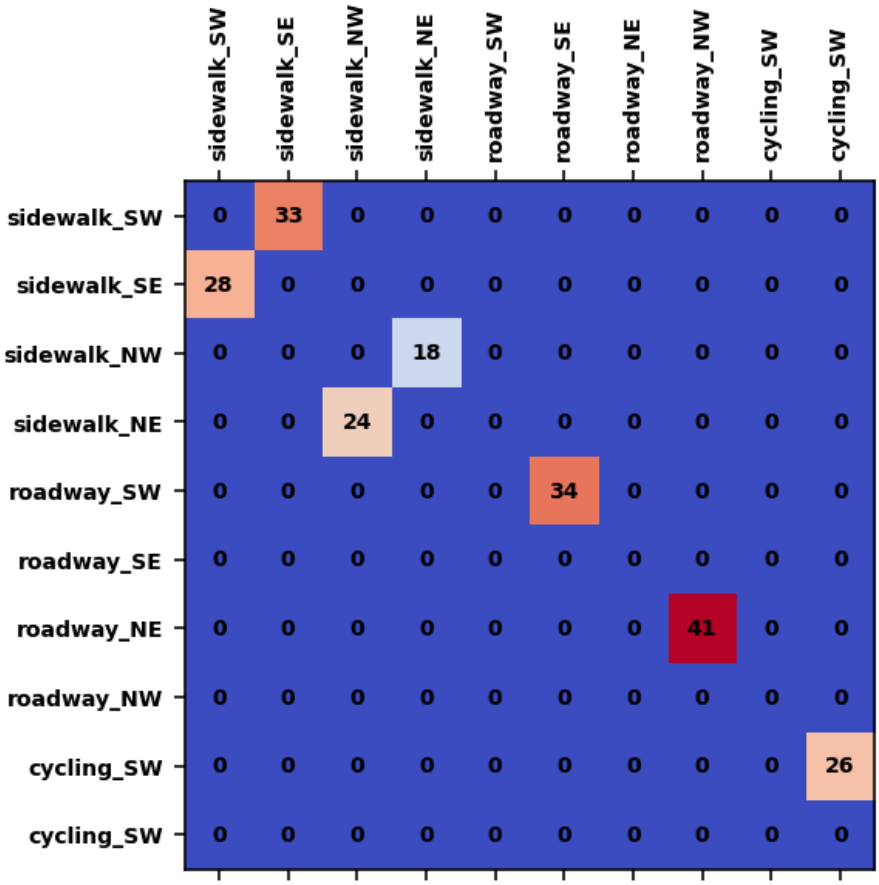

Temporal distribution of activities by activityType
Monday, Sep 14, 2020
strolling
gogging
shopping
talking
resting
eating
playing
doing_exercise
smoking
using_cellphone
observing
No. of activities
0
2
4
6
8
10
08:00
10:00
12:00
14:00
16:00
18:00
20:00
Time



OD matrix for observed pedestrians

The above OD matrix shows the following indicators:

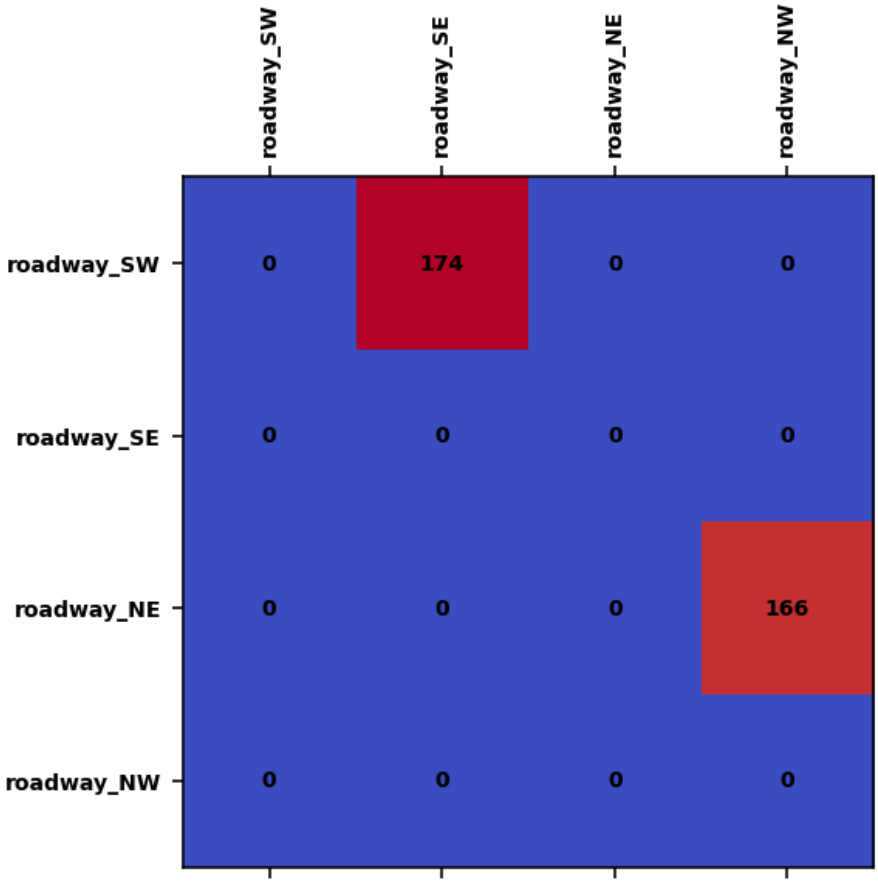
- The number of people passing in each direction of sidewalks
- The number of people leaving each ZOI
    - 10 people left the Richmond building
- The direction of people after leaving a ZOI:
    - 10 people headed to sidewalk_NW (#103 in the Figure 1) after leaving Richmond building
- The number of people crossing the street:
    - None of people crossed the street (for example: departing from Richmond Bldg. heading to McDonalds- See Figure 1)
- The number of people entering to a ZOI:
    - 14 people entered to Sam store
- The most active ZOI:
    - Richmond (residential) Bldg is the most active ZOI



OD matrix for observed cyclists

The above OD matrix shows the following indicators:

- The number of cyclists using cycling path:
  - 26 cyclists used cycling path
- The number of cyclists riding in sidewalks
  - 33 cyclists rode on the southern sidewalk (from #101 to #102 in the Figure 1)
- The number of cyclists riding on the roadways
  - 41 cyclists rode on the northern roadway (from #203 to #204)
- The number of cyclists riding against the traffic flow
  - None of the cyclists rode from #204 to #203 or from #202 to #201
- The number of cyclists cross the street
  - None of the cyclist rode from #301 to #104



OD matrix for observed cyclists

The above OD matrix shows the following indicators:

- The number of passing vehicles in each direction (lanes):
    - 174 vehicles passed from #201 to #202
- The number of vehicles that do U-turn
    - None of the vehicle do U-turn (for example from #201 to #204)